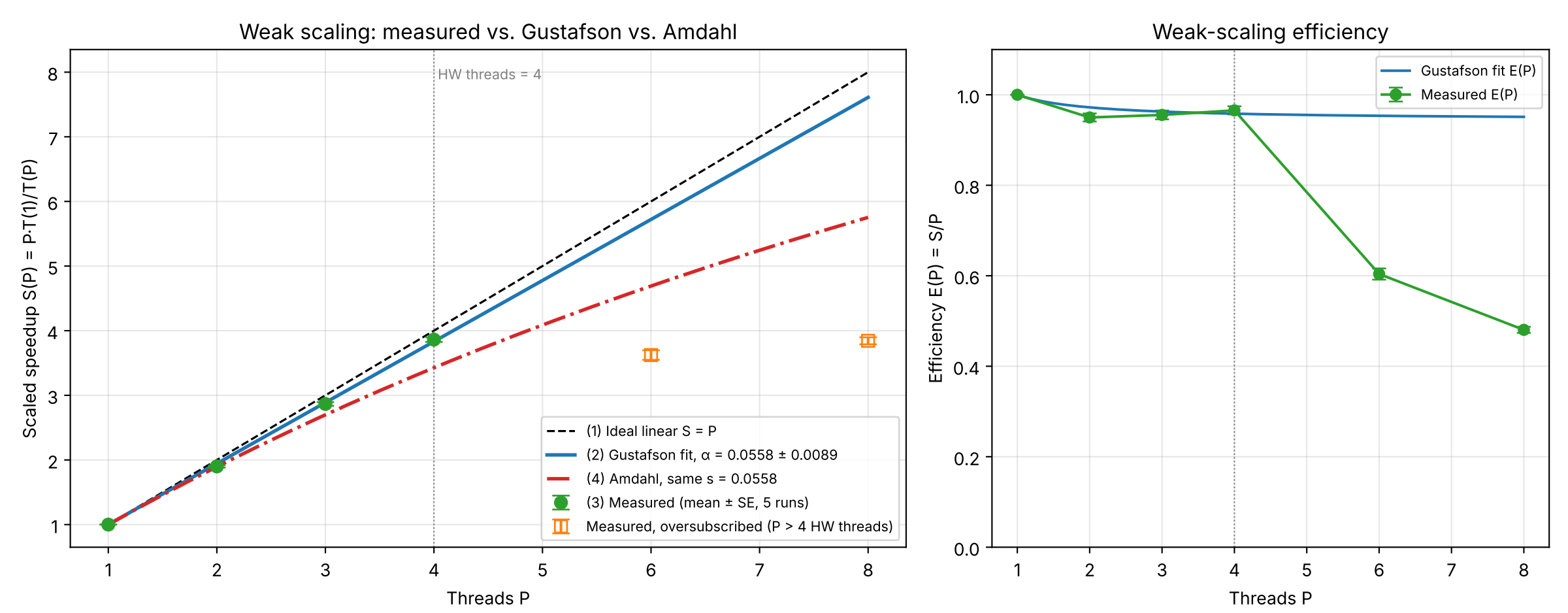
Weak scaling: measured vs. Gustafson vs. Amdahl
HW threads = 4
Scaled speedup S(P) = P·T(1)/T(P)
Threads P
(1) Ideal linear S = P
(2) Gustafson fit, α = 0.0558 ± 0.0089
(4) Amdahl, same s = 0.0558
(3) Measured (mean ± SE, 5 runs)
Measured, oversubscribed (P > 4 HW threads)
Weak-scaling efficiency
Efficiency E(P) = S/P
Threads P
Gustafson fit E(P)
Measured E(P)